X. Lab 10: Flip-flops and sequential logic

A. Introduction

There are two categories of logic. *Combinational logic* is logic in which the state of the logic circuit is determined only by its inputs. *Sequential logic* is logic in which the state of the logic circuit depends on its input *and* its past history.

1. SR flip-flop

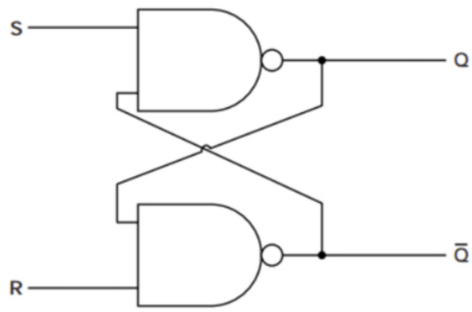
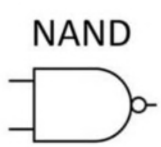


Figure X-1. SR flip-flop

The simplest sequential logic circuit is the *Set-Reset (SR) flip-flop*. It can be constructed from two NAND gates as shown in Figure I-1. To understand how this circuit works, first recall that the NAND truth table is as given in Figure X-2. Now, first suppose that $S = R = 1$ and that output $Q = 0.$ In that case it must be that $\bar{Q} = 1,$ because $\bar{Q} = 0$ only if both inputs to the lower NAND gate are 1. But now if $\bar{Q} = 1$, it must be that $Q = 0,$ since both inputs to the upper NAND gate are 1. This consistent with our initial assumption for output *Q*, so it follows that { $R = S = 1$, $Q = 0,$ $\bar{Q} = 1$ } is a stable state of the circuit.



A	B	Output
0	0	1
1	0	1
0	1	1
1	1	0

Figure X-2. NAND gate truth table.

On the other hand, suppose that $R = S = 1$ and $Q = 1.$ It follows that $\bar{Q} = 0$ from which it follows that $Q = 1.$ So $R = S = 1$, $Q = 1,$ and $\bar{Q} = 0$ is *also* a stable state of the circuit. Since there are two stable states of the circuit, its state at any given time depends not only on its inputs *R* and *S*, but also on its past history.

"set" function: Suppose that $S=0,\ R=1 \ \Rightarrow\ \bar{Q}=0,\ Q=1$.

If now $S \to 1$, the output $\bar{Q}=0,\ Q=1$ remains unchanged. Thus, $S=0$, $R=1$ is an input that "sets" the output to $\bar{Q}=0,\ Q=1$, where it remains as long as $R=1$.

"reset" function: Suppose that $S=1,\ R=0 \ \Rightarrow\ \bar{Q}=1,\ Q=0$.

If now $S \to 0$, the output $\bar{Q}=1,\ Q=0$ remains unchanged. Thus, $S=1$, $R=0$ is an input that "resets" the output to $\bar{Q}=1,\ Q=0$, where it remains as long as $R=0$.

We can summarize this with the truth table shown in Table X-1.

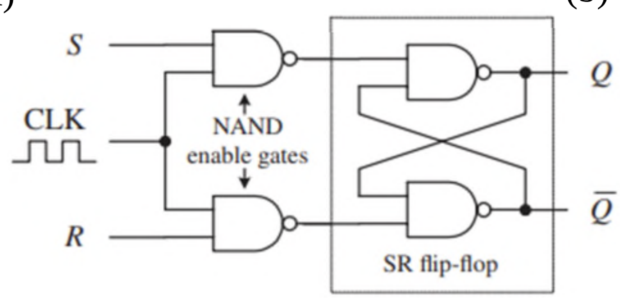
S_{n+1}	R_{n+1}	Q_{n+1}	$\bar{Q}_{n+1}$
0	0	1 (not used)	1 (not used)
1	0	0	1
0	1	1	0
1	1	Q_n	$\bar{Q}_n$

Table X-1. SR flip-flop truth table

In this table, we imagine that we have a series of inputs to the flip-flop $(\{S_1,R_1\},\{S_2,R_2\},\ldots\{S_n,R_n\},\{S_{n+1},R_{n+1}\},\ldots)$. The notation in the last row indicates that the circuit outputs Q_{n+1} and $\bar{Q}_{n+1}$ are unchanged from their previous values Q_n and $\bar{Q}_n$.

2. Clocked SR flip-flop

(a)



(b)

CLK	S	R	Q	$\bar{Q}$	Mode	
0	0	0	Q	$\bar{Q}$	hold	
0	0	1	Q	$\bar{Q}$	hold	*SR*-inputs
0	1	0	Q	$\bar{Q}$	hold	disabled
0	1	1	Q	$\bar{Q}$	hold	
1	0	0	0	0	hold	
1	0	1	0	1	RESET	*SR*-inputs
1	1	0	1	0	SET	enabled
1	1	1	Q	$\bar{Q}$	indeterminate	

Figure X-3. (a) Clocked SR flip-flop (b) Truth table for the clocked SR flip-flop. Figure is from PEFI, 4th ed.

It is sometimes useful to have a flip-flop which changes state only when enabled by a control or "clock" input. Flip flops that respond this way are referred to as *synchronous* or *clocked* flip-flops. To do this, we just need to attach enable gates to each of the inputs of the flip-flops, as shown in Figure X-3(a). This circuit has the truth table shown in Figure X-3(b). The differences that this circuit has from the asynchronous flip-flop of Figure X-1 are that (i) if CLK is low, the circuit holds its output regardless of what *R* and *S* do, (ii) if CLK is high, the circuit output for a given *R* and *S* is the that of the asynchronous flip-flip with input $\bar{R}$ and $\bar{S}$ (for instance reset is now $S=0,\ R=1$), and (iii) there is a state with indeterminate output which should be avoided.

3. Debounced switches

Ordinary switches suffer from *switch bounce*. This means that they don't cleanly switch on and off. Rather, if you're turning the switch on, the switch will bounce between on and off a few times before it settles in at on. Similarly it can bounce when you turn it off. This creates problems when using sequential logic, because the state of your logic circuit depends on the switch bounce, which is indefinite. This is illustrated in Figure X-4.

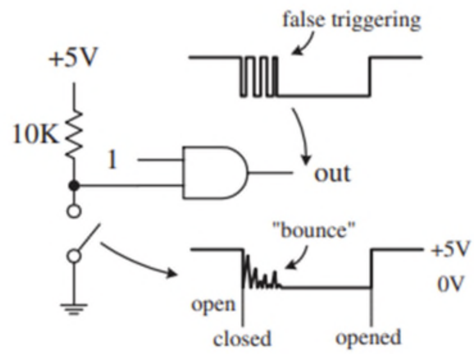


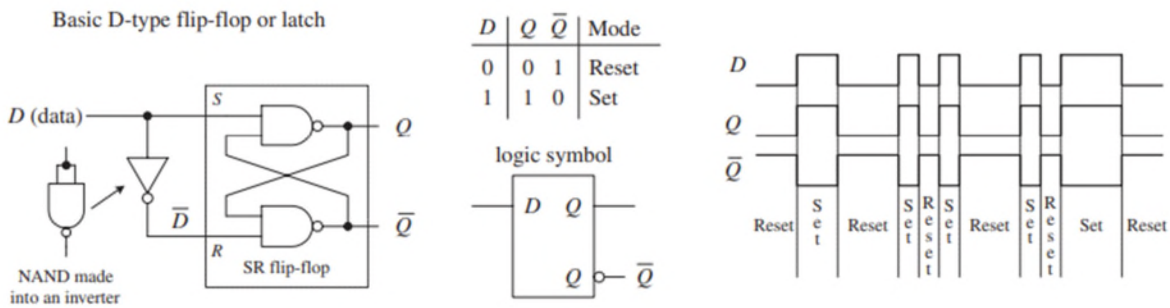
Figure X-4. Switch bounce effect on a logic circuit. Figure is from PEFI, 4th ed.

To solve this problem, *debounced switches* are used. These have additional circuitry to eliminate the bounce, so that the switch cleanly makes a single transition from off to on or from on to off. Your Global Specialties protoboards have two debounced switches on them, and you should use these to produce any digital input where switch bounce could create problems.

4. Type D flip-flop

This section is copied from PEFI, 4th ed.

A D-type flip-flop (data flip-flop) is a single input device. It is basically an SR flip-flop, where *S* is replaced with *D* and *R* is replaced with $\bar{D}$ (inverted *D*). The inverted input is tapped from the *D* input through an inverter to the *R* input, as shown in Fig. 12.63. The inverter ensures that the indeterminate condition (race, or not used state, $S = 1$, $R = 1$) never occurs. At the same time, the inverter eliminates the hold

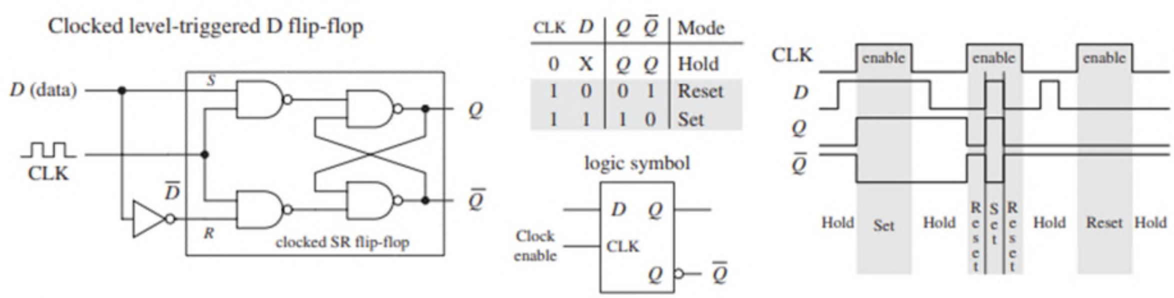


D	Q	$\overline{Q}$	Mode
0	0	1	Reset
1	1	0	Set

FIGURE 12.63

condition so that you are left with only set ($D = 1$) and reset ($D = 0$) conditions. The circuit in Fig. 12.63 represents a level-triggered D-type flip-flop.

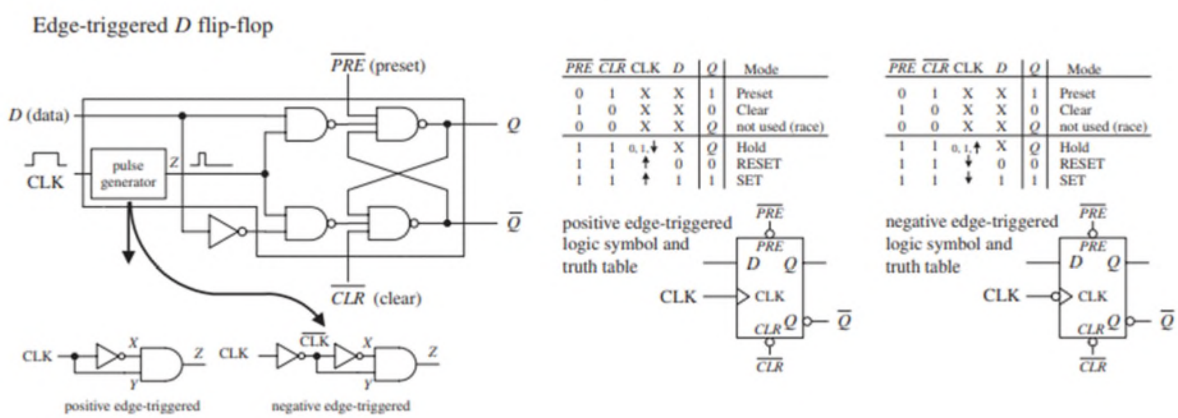
To create a clocked D-type level-triggered flip-flop, first start with the clocked level-triggered SR flip-flop and throw in the inverter, as shown in Fig. 12.64.



CLK	D	Q	$\overline{Q}$	Mode
0	X	Q	$\overline{Q}$	Hold
1	0	0	1	Reset
1	1	1	0	Set

FIGURE 12.64

To create a clocked, edge-triggered, D-type flip-flop, take a clocked edge-triggered SR flip-flop and add an inverter, as shown in Fig. 12.65.



$\overline{PRE}$	$\overline{CLR}$	CLK	D	Q	Mode
0	1	X	X	1	Preset
1	0	X	X	0	Clear
0	0	X	X	Q	not used (race)
1	1	0, 1, ↓	X	Q	Hold
1	1	↑	0	0	RESET
1	1	↑	1	1	SET

$\overline{PRE}$	$\overline{CLR}$	CLK	D	Q	Mode
0	1	X	X	1	Preset
1	0	X	X	0	Clear
0	0	X	X	Q	not used (race)
1	1	0, 1, ↑	X	Q	Hold
1	1	↓	0	0	RESET
1	1	↓	1	1	SET

FIGURE 12.65

Figure 12.66 shows a popular edge-triggered D-type flip-flop IC, the 7474 (for example, the 74HC74). It contains two D-type positive edge-triggered flip-flops with asynchronous preset and clear inputs.

74HC74 Dual D-type positive edge-triggered flip-flop with preset and clear

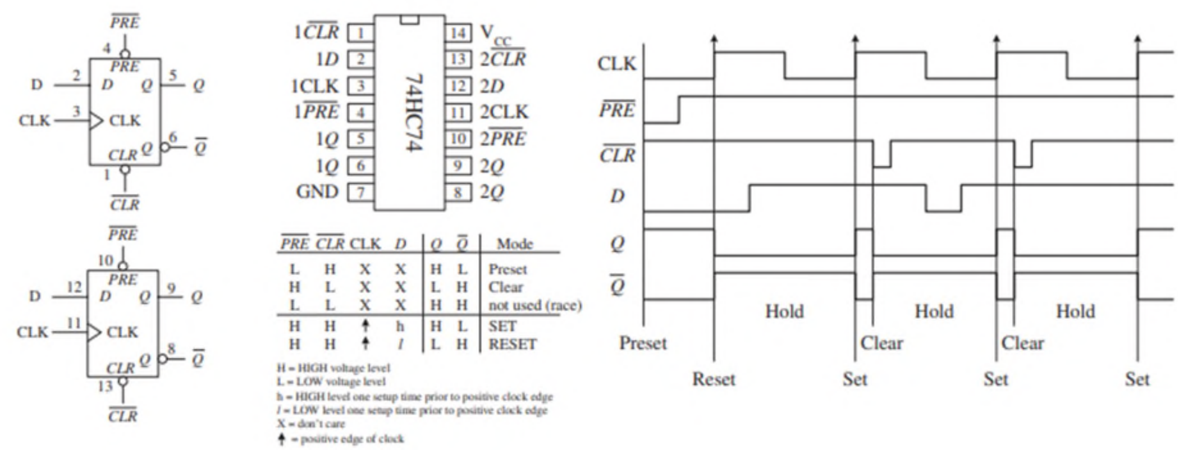


FIGURE 12.66

Note the lowercase letters *l* and *h* in the truth table in this figure. The *h* is similar to the *H* for a high voltage level, and the *l* is similar to the *L* for low voltage level; however, there is an additional condition that must be met for the flip-flop's output to do what the truth table indicates. The additional condition is that the *D* input must be fixed high (or low) in duration for at least one *setup time* (t_s) before the positive clock edge. This condition stems from the real-life propagation delays present in flip-flop ICs. If you try to make the flip-flop switch states too fast (do not give it time to move electrons around), you can end up with inaccurate output readings. For the 7474, the setup time is 20 ns. Therefore, when using this IC, you must not apply input pulses that are within the 20-ns limit. Other flip-flops will have different setup times, so you will need to check the manufacturer's data sheets. We will discuss setup time and some other flip-flop timing parameters in greater detail in Sec. 12.6.6.

B. Procedure

For this lab I'll skip explicit directions "For your report." When you prepare your report, just provide detailed descriptions of your measurements and results and answers to the questions raised as you normally would.

Note that as before, use only +5V and GND for power connections, and be sure that you're using a good +5 V supply. (The ones on the PB505A are not working.) The CMOS ICs burn out easily and are sensitive to static electricity. Let your TA know if you're having trouble with burned out ICs. Make all circuit connections including power and ground with the power off and only turn the power on when you have all the connections in place.

1. Flip-flops from NAND gates

a) <u>Asynchronous SR flip-flip</u>

Using two NAND gates, built the circuit shown in Figure X-1. Verify the functions of the R and S inputs. You should report the sequence of R and S values that you used along with outputs Q and $\bar{Q}$, and this sequence should include enough steps to fully test the operation of the flip-flip including its ability to hold the output for certain changes in R and S.

b) Clocked SR flip-flop

Nest, use two more NAND gates to add the clock to your flip-flop as shown in Figure X-3(a). Verify the functions of the R, S, and CLK inputs. You should report the sequency of R, S, and CLK values that you used along with outputs Q and $\overline{Q}$, and your sequence should include enough steps to fully test the operation of the flip-flop.

2. D-type flip-flop

For the next several exercises you will use the SN74HCT74N dual D-type edge-triggered flip flop. Some specifications for this IC are given at the end of this chapter.

a) D-type flop-flop operation

Tie the SET (called PRESET) and RESET (Called CLEAR) pins to +5V (HI) as shown in Figure X-5. Connect the D-input to a slide switch on your breadboard and clock the input with a debounced pushbutton switch. The "normally open" switch connection needs a pull-up resistor to +5V. Tie the inputs for the second flip-flop (the one you're not using) HI. Check the operation of the flip-flip by verifying that the D-input appears on the Q-output when and only when clocked. Assert (pull low) SET and report what happens. Assert (pull low) RESET and report what happens.

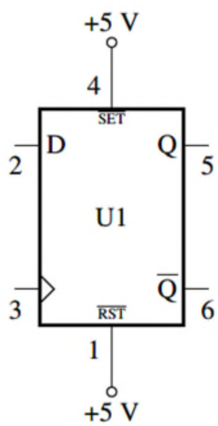


Figure X-5. D-type flip-flop connections

b) Flip-flop with feedback

Now apply some feedback as shown in Figure X-6. Clock the circuit manually and observe the output. Then connect the clock input to the TTL output of a function generator. Watch the Clock input and the Q output on an oscilloscope. What is the circuit doing? Now turn the frequency up and measure the *propagation delay* of the circuit. Why is this relevant given the feedback connection from $\overline{Q}$ to D?

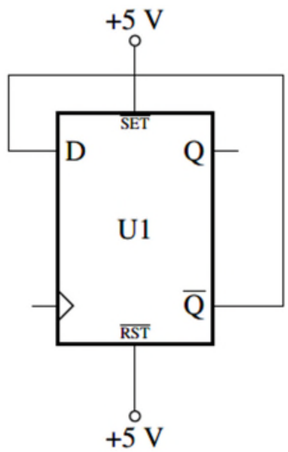


Figure X-6. D-type flip-flop with feedback.

c) Cascaded flip-flops

Now cascade two flip-flops as shown in Figure X-7. Keep the SET and RESET inputs tied to HI (omitted from the figure for clarity). Using the function generator as the clock input for U1, observe and record the Q output of U2. This is called a ripple counter because of the way the second flip-flop is clocked.

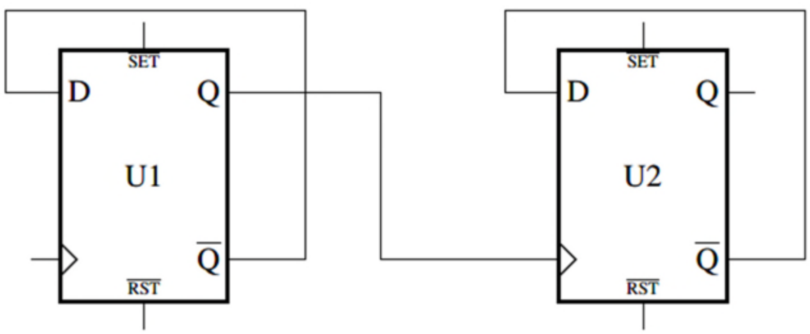


Figure X-7. Cascade of two flip-flops with feedback.

d) Shift register

Build the shift register shown in Figure X-8. (You will need a second 74HCT74.) Use a function generator to provide the clock and a second function generator to generate the IN pulses. Make the IN pulses at least ten times slower than the clock. Using the scope, observe IN and Q output of the first flip-flop. Why the jitter?

Next, watch the output Q_2 along with IN. Repeat for Q_3 and Q_4. What is the circuit doing?

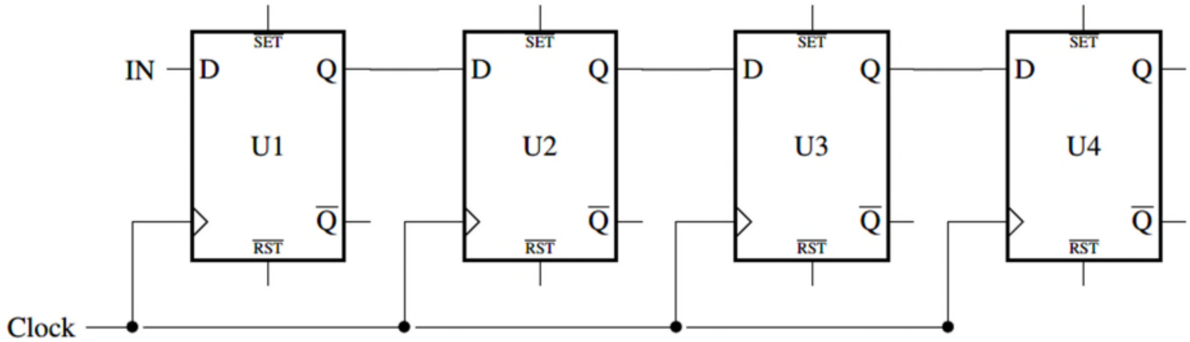


Figure X-8. A shift register made with four D-type flip-flops.

C. IC Specifications

SNx4HCT74 Dual D-Type Positive-Edge-Triggered Flip-Flips With Clear and Preset

1 Features

- Operating voltage range of 4.5 V to 5.5 V
- Outputs can drive up to 10 LSTTL loads
- Low power consumption, 40-μA max I_{CC}
- Typical t_{pd} = 17 ns
- ±4-mA output drive at 5 V
- Low input current of 1 μA max
- Inputs are TTL-voltage compatible

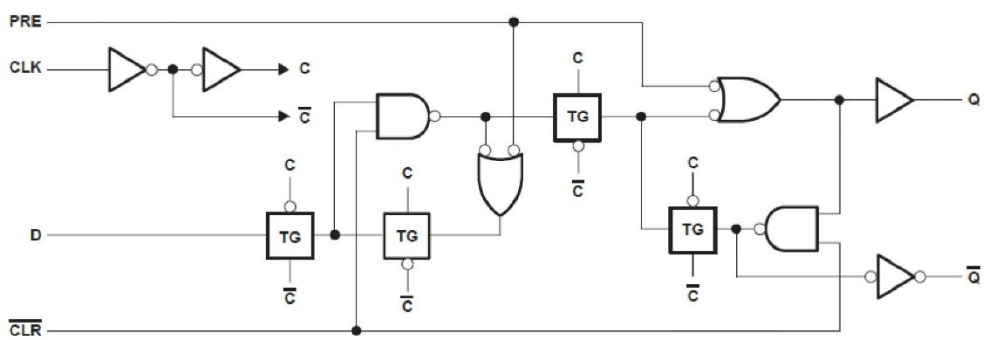
2 Description

The 'HCT74 devices contain two independent D-type positive-edge-triggered flip-flops. A low level at the preset ($\overline{PRE}$) or clear ($\overline{CLR}$) inputs sets or resets the outputs, regardless of the levels of the other inputs. When $\overline{PRE}$ and $\overline{CLR}$ are inactive (high), data at the data (D) input meeting the setup time requirements are transferred to the outputs on the positive-going edge of the clock (CLK) pulse. Clock triggering occurs at a voltage level and is not directly related to the rise time of CLK. Following the hold-time interval, data at the D input may be changed without affecting the levels at the outputs.

Device Information

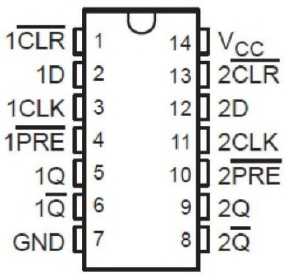
PART NUMBER	PACKAGE(1)	BODY SIZE (NOM)
SN74HCT74D	SOIC (14)	8.65 mm × 3.90 mm
SN74HCT74DB	SSOP (14)	6.20 mm × 5.30 mm
SN74HCT74N	PDIP (14)	19.31 mm × 6.35 mm
SN74HCT74NS	SO (14)	10.20 mm × 5.30 mm
SN74HCT74PW	TSSOP (14)	5.00 mm × 4.40 mm
SNJ54HCT74FK	LCCC (20)	8.89 mm × 8.45 mm
SNJ54HCT74W	CFP (14)	9.21 mm × 6.29 mm
SNJ54HCT74J	CDIP (14)	19.55 mm × 6.71 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

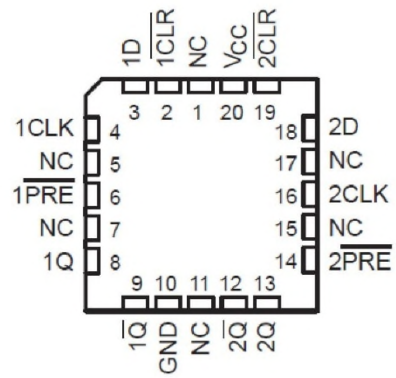


Functional Block Diagram

4 Pin Configuration and Functions



J, W, D, DB, N, NS, or PW package
14-Pin CDIP, CFP, SOIC, SSOP, PDIP, SO, or TSSOP
Top View



NC - No internal connection

FK Package
20-Pin LCCC
Top View

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage range		–0.5	7	V
I_{IK}	Input clamp current(2)	$V_I < 0$ or $V_I > V_{CC}$		± 20	mA
I_{OK}	Output clamp current(2)	$V_O < 0$ or $V_O > V_{CC}$		± 20	mA
I_O	Continuous output current	$V_O = 0$ to V_{CC}		± 25	mA
	Continuous current through V_{CC} or GND			± 50	mA
T_J	Junction temperature			150	°C
T_{stg}	Storage temperature range		–65	150	°C

(1) Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

5.2 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)(1)

			SN54HCT74(2)			SN74HCT74			UNIT
			MIN	NOM	MAX	MIN	NOM	MAX	
V_{CC}	Supply voltage		4.5	5	5.5	4.5	5	5.5	V
V_{IH}	High-level input voltage	V_{cc} = 4.5 V to 5.5 V	2			2			V
V_{IL}	Low-level input voltage	V_{cc} = 4.5 V to 5.5 V			0.8			0.8	V
V_I	Input voltage		0		V_{CC}	0		V_{CC}	V
V_O	Output voltage		0		V_{CC}	0		V_{CC}	V
Δt/Δv	Input transition rise/fall time				500			500	ns
T_A	Operating free-air temperature		−55		125	−40		85	°C

(1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.
(2) SN54HCT74 is in product preview.

5.4 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		V_{CC}	T_A = 25°C			SN54HCT74[(2)]		SN74HCT74		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
V_{OH}	$V_I = V_{IH}$ or V_{IL}	I_{OH} = −20 µA	4.5 V	4.4	4.499		4.4		4.4		V
		I_{OH} = −4 mA		3.98	4.3		3.7		3.84		
V_{OL}	$V_I = V_{IH}$ or V_{IL}	I_{OL} = 20 µA	4.5 V		0.001	0.1		0.1		0.1	V
		I_{OL} = 4 mA			0.17	0.26		0.4		0.33	
I_I	$V_I = V_{CC}$ or 0		5.5 V		±0.1	±100		±1000		±1000	nA
I_{CC}	$V_I = V_{CC}$ or 0,	I_O = 0	5.5 V			4		80		40	µA
ΔI_{CC} [(1)]	One input at 0.5 V or 2.4 V, Other inputs at 0 or V_{CC}		5.5 V		1.4	2.4		3		2.9	mA
C_i			4.5 V to 5.5 V		3	10		10		10	pF

(1) This is the increase in supply current for each input that is at one of the specified TTL voltage levels, rather than 0 V or V_{CC}.
(2) SN54HCT74 is in product preview.

5.5 Timing Requirements

overrecommended operating free-air temperature range (unless otherwise noted)

			V_{CC}	T_A = 25°C		SN54HCT74[(1)]		SN74HCT74		UNIT
				MIN	MAX	MIN	MAX	MIN	MAX	
f_{clock}	Clock frequency		4.5 V		27		18		22	MHz
			5.5 V		30		20		24	
t_w	Pulse duration	$\overline{PRE}$ or $\overline{CLR}$ low	4.5 V	16		24		20		ns
			5.5 V	14		21		18		
		CLK high or low	4.5 V	18		27		23		
			5.5 V	16		24		21		
t_{su}	Setup time before CLK↑	Data	4.5 V	12		18		15		ns
			5.5 V	11		16		14		
		$\overline{PRE}$ or $\overline{CLR}$ inactive	4.5 V	0		0		0		
			5.5 V	0		0		0		
t_h	Hold time, data after CLK↑		4.5 V	0		0		0		ns
			5.5 V	0		0		0		

(1) SN54HCT74 is in product preview.

5.6 Switching Characteristics

over recommended operating free-air temperature range, C_L = 50 pF (unless otherwise noted) (see Parameter Measurement Information)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CC}	T_A = 25°C			SN54HCT74[(1)]		SN74HCT74		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
f_{max}			4.5 V	27	40		18		22		MHz
			5.5 V	30	46		20		24		
t_{pd}	$\overline{PRE}$ or $\overline{CLR}$	Q or $\overline{Q}$	4.5 V		21	35		53		44	ns
			5.5 V		17	31		48		40	
	CLK	Q or $\overline{Q}$	4.5 V		20	28		42		35	
			5.5 V		18	25		38		31	
t_t		Q or $\overline{Q}$	4.5 V		8	15		22		19	ns
			5.5 V		7	14		20		17	

(1) SN54HCT74 is in product preview.

5.7 Operating Characteristics

T_A = 25°C

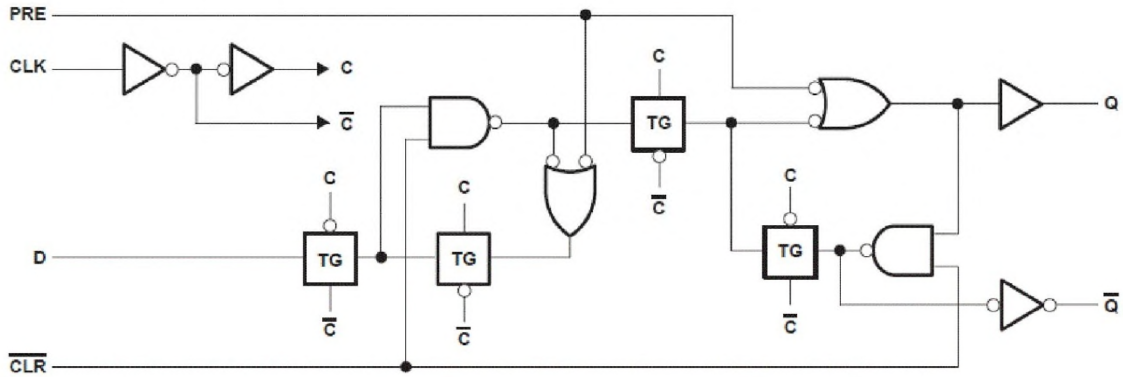
PARAMETER		TEST CONDITIONS	TYP	UNIT
C_{pd}	Power dissipation capacitance per flip-flop	No load	35	pF

7 Detailed Description

7.1 Overview

The 'HCT74 devices contain two independent D-type positive-edge-triggered flip-flops. A low level at the preset ($\overline{PRE}$) or clear ($\overline{CLR}$) inputs sets or resets the outputs, regardless of the levels of the other inputs. When $\overline{PRE}$ and $\overline{CLR}$ are inactive (high), data at the data (D) input meeting the setup time requirements are transferred to the outputs on the positive-going edge of the clock (CLK) pulse. Clock triggering occurs at a voltage level and is not directly related to the rise time of CLK. Following the hold-time interval, data at the D input may be changed without affecting the levels at the outputs.

7.2 Functional Block Diagram



7.3 Device Functional Modes

Table 7-1. Function Table

INPUTS				OUTPUT	
$\overline{PRE}$	$\overline{CLR}$	CLK	D	Q	$\overline{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H(1)	H(1)
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q_0	$\overline{Q}_0$

(1) This configuration is nonstable; that is, it does not persist when $\overline{PRE}$ or $\overline{CLR}$ returns to its inactive (high) level.